Non-Iterative Enhanced SDP Relaxations for Optimal Scheduling of Distributed Energy Storage in Distribution Systems

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***Abstract*—Convexification of an optimal scheduling algorithm for distributed energy storage (DES) in radial distribution systems with high penetration of photovoltaic resources is studied. The AC power flow equalities are taken into account as constraints in the optimization model. Different from the typical optimal power flow problem, the objective function of a DES optimal scheduling (DESOS) problem varies with changing operational requirements. In this paper, three frequently-used objective functions are considered for the DESOS problem. Two of them are monotonic over the feasible set while the third is not. An illustrative example elucidates that the descent direction of a chosen objective function significantly impacts the efficiency of the second-order cone programming (SOCP) relaxation for the DESOS problem. To obtain tighter semidefinite programming (SDP) relaxations for the DESOS cases where the SOCP relaxation is not exact, this paper looks for computationally efficient convex constraints that can approximate the rank-1 constraint in the non-iterative framework. The designed non-iterative enhanced SDP relaxations are compared in terms of tightness of convexification for the DESOS problems considering the three objective functions independently. The comparison is performed on several radial IEEE test systems and a real world distribution feeder.**

***Index Terms*—Convex relaxation, distributed energy storage (DES), enhanced SDP (ESDP) relaxation, non-iterative, optimal dispatch.**

I. Introduction

SINCE distributed energy storage (DES) has capabilities of both time-shifting of energy supplied and fast-ramping [1], it has drawn a great deal of attentions in facilitating the integration of high penetration of renewable generation resources [2]–[5]. This paper studies an optimal dispatch algorithm for DES in distribution systems with integration of high penetration of photovoltaic (PV) resources. In distribution systems, the popular linear approximation of power flow equations, the DC power flow model [6], is no longer suitable for the DES optimization problem due to the high r/x-ratio of the feeders. However, the conventional AC power flow equations impose challenges in solving the DES optimal scheduling (DESOS) problem since they are non-convex. The non-convexity of the conventional AC power flow equations results in the DES optimization problems being non-deterministic polynomial-time hard (NP-hard) to solve and only guarantees a locally optimal solution. The NP-hardness in general networks and radial networks has been studied in [7] and [8] respectively.

Recently, a number of publications have studied the second order cone programming (SOCP) relaxation as well as the semidefinite programming (SDP) relaxation for the optimal power flow (OPF) algorithms (refer to [7] and the references therein). In OPF problems, researchers usually consider an objective function which is a linear or a convex nonlinear function in terms of the outputs of the generators and represents the generation cost. However, the objective function of a DESOS problem may vary due to the needs for DES to meet various operational requirements. A preliminary discussion in [5] points out that, in addition to tightness, the descent direction of an objective function [9] in the feasible set plays an important role in the exactness of the SOCP relaxation. Three frequently-used objective functions are considered for the DESOS algorithm (see Section II-A). The considered objective functions reflect actual operating requirements in distribution systems. Compared with the conventional OPF problem, the additional objective functions and DES constraints may significantly increase the number of factors that can affect the exactness of convex relaxations. To intuitively assess the impacts of the chosen objective functions on the exactness of the SOCP relaxation, the feasible sets of a 2-bus system are plotted to facilitate a theoretical discussion. A numerical example on an actual feeder also shows that the dynamic constraint of DES which is linear may also affect the exactness of the SOCP relaxation.

Since the SOCP relaxation performs unsatisfactorily for a number of DESOS cases, the SDP relaxation is employed to convexify the DESOS algorithm. However, it has been proven in [10, Proposition 3.4] that the SOCP relaxation for the AC power flow equalities in radial networks described by branch flow models (BFMs) in [11] and [12] is tighter than the basic form of the SDP relaxation. Hence, it is valuable to explore tighter SDP relaxations for these DESOS cases. In [7], [13] and [14], the authors give insights into the properties of the SOCP relaxation and the basic SDP (BSDP) relaxation of the generic OPF problem. However, methods to tighten the convex relaxations are not well discussed.

Manuscript received November 5, 2015; revised February 21, 2016, May 4, 2016, and June 21, 2016; accepted July 23, 2016. Date of publication July 27, 2016; date of current version April 17, 2017. This work was supported by the U.S. Department of Energy under Grant DE-EE0004679. Paper no. TPWRS-01576-2015.

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Digital Object Identifier 10.1109/TPWRS.2016.2594298

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Reference [15] applied a penalty method to the SDP relaxation to enforce the rank of its solution to become nearly 1 iteratively based on a premise that the rank of solution for the BSDP relaxation is low. Valid linear cuts are generated for tightening the SOCP relaxation for OPFs in an iterative framework in [16]. Interesting results have been reported with these iterative methods. However, an iterative procedure usually needs to solve a problem repeatedly, which increases the computational burden. Moreover, the convergence of most of these iterative methods is not well studied. As an alternative, this paper looks for computationally effective convex constraints that are valid for tightening the BSDP relaxation of the DESOSs in the non-iterative framework. In reference [17], non-iterative linear cuts are proposed based on the SOCP formulation of OPF to provide a tight root node for solving the relaxed OPF problem in BARON [18].

The objective of this paper is: 1) assess the impacts of the chosen objective functions on the exactness of the convex relaxations for the DESOS problem; 2) enhance the SDP relaxation with different non-iterative convex constraints for DESOS problems in radial networks. The rest of the paper is organized as follow: the DESOS problems in radial networks are modeled and their BSDP relaxation is introduced in Section II. In Section III, the impact of objective functions on the exactness of the convex relaxations for DESOS is discussed through an illustrative example. Several convex constraints proposed in the analytical optimization literature are introduced to construct non-iterative enhanced SDP (ESDP) relaxations for the DESOS problem and a numerical comparison of the ESDP relaxations is provided in Sections IV and V respectively. Conclusions are made in Section VI.

II. Modeling of DESOS and Its BSDP Relaxation

A. Modeling of the DESOS

The topology of distribution systems is usually radial. Hence, a BFM is used to describe the AC power flow in radial networks in the DESOS model since the BFMs are exact for radial networks and contain fewer non-convex constraints [5]. The BFMs proposed in [11] and [12] are equivalent since there is a bijection between them. Using the BFM proposed in [11], the DESOS problem is modeled as follows:

$$\min\ f_1 = \sum_t^T c_t p_t^{\mathrm{Grid}} \tag{1a}$$

$$\min\ f_2 = \sum_t^T \left(\sum_{ij}^M r_{ij}\ell_{ij,t} + \sum_i^{N_T} k_i v_{i,t} + \sum_i^{N_S} p_{i,t}^{\mathrm{DESloss}} \right) \tag{1b}$$

$$\min\ f_3 = \sum_t^T \left(\sum_i^N \left| v_{i,t} - v_i^{\mathrm{set}} \right| \right) \tag{1c}$$

$$\begin{aligned} &\text{s.t. } p_{i,t}^{\mathrm{DCH}} + p_{i,t}^{\mathrm{CH}} + p_{(i=1),t}^{\mathrm{Grid}} + p_{i,t}^{\mathrm{PV}} \\ &- p_{i,t}^{\mathrm{Load}} - k_{(i\in N_T)} v_{(i\in N_T),t} \\ &= \sum_{k:i\to k} P_{ik,t} - (P_{ji,t} - r_{ij}\ell_{ji,t}) \end{aligned} \tag{2a}$$

TABLE I
Nomenclature in Section II

SYMBOL	Dimension/quantity
c_t	Cost of grid energy ($/MWh). In a market environment, it represents the NP at the distribution substation location.
$E_i^{\mathrm{DES}}, E_i^{\mathrm{DES}}$	Energy rating and surplus of the DES at bus *i*
K_i, N_T	Loss coefficient and high voltage bus set of transformers
M, N, Ns	Branch set, total bus set and DES bus set respectively
$p_{i,t}^{\mathrm{DCH}}, p_{i,t}^{\mathrm{CH}}$	Discharging and charging rate of DES at bus *i* at time *t* respectively
$p_{(i=1),t}^{\mathrm{Grid}}, p_{(i=1),t}^{\mathrm{Grid}}$	Active and reactive grid power at hour *t*
P_i^{DES}	Power rating of the DES at bus *i*
$p_{i,t}^{\mathrm{DESloss}}$	Power losses of DES at bus *i* at hour *t*
$p_{i,t}^{\mathrm{Load}}, q_{i,t}^{\mathrm{Load}}$	Active and reactive load at bus *i* at hour *t*
$p_{i,t}^{\mathrm{PV}}$	PV generation at bus *i* and at hour *t*
$P_{ij,t}, Q_{ij,t}, \ell_{ij,t}$	Active, reactive power flow and square of current magnitude in branch *ij* at hour *t*
r_{ij}, x_{ij}	Resistance and reactance of branch *ij*
R	Substation MVA rating
$\bar{S}_{ij}$	Thermal limit of line *ij*
$v_{i,t}, v_i^{\mathrm{set}}$	Square of voltage magnitude at bus *i* at hour *t* and its set point
$\ell_{ij,t}$	Square of magnitude of current in branch *ij* at hour *t*
η_c, η_d	Charging/discharging efficiency respectively
T, Δt	Operation cycle and time interval between two operations of the DES respectively

$$q_{(i=1),t}^{\mathrm{Grid}} - q_{i,t}^{\mathrm{Load}} = \sum_{k:i\to k} Q_{ik,t} - (Q_{ji,t} - x_{ij}\ell_{ji,t}) \tag{2b}$$

$$v_{i,t} = v_{j,t} + 2\left(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}\right) - \left(r_{ij}^2 + x_{ij}^2\right)\ell_{ij,t} \tag{2c}$$

$$v_{i,t}\ell_{ij,t} = P_{ij,t}^2 + Q_{ij,t}^2 \tag{2d}$$

$$0 \le E_i^{\mathrm{spl}} - \sum_{t=1}^{t'} \left(\eta_c p_{i,t}^{\mathrm{CH}} + p_{i,t}^{\mathrm{DCH}} / \eta_d\right)\Delta t \le E_i^{\mathrm{DES}} \tag{3a}$$

$$-P_i^{\mathrm{DES}} \le p_{i,t}^{\mathrm{CH}} \le 0 \tag{3b}$$

$$0 \le p_{i,t}^{\mathrm{DCH}} \le P_i^{\mathrm{DES}} \tag{3c}$$

$$p_{i,t}^{\mathrm{DESloss}} = \left(1/\eta_d - 1\right)p_{i,t}^{\mathrm{DCH}} + \left(\eta_c - 1\right)p_{i,t}^{\mathrm{CH}} \tag{3d}$$

$$p_{i,t}^{\mathrm{CH}} p_{i,t}^{\mathrm{DCH}} = 0 \tag{3e}$$

$$P_{ij,t}^2 + Q_{ij,t}^2 \le \bar{S}_{ij}^2 \tag{4a}$$

$$0 \le \ell_{ij,t} \le \bar{\ell}_{ij} \tag{4b}$$

$$\underline{v_i} \le v_{i,t} \le \bar{v}_i \tag{4c}$$

$$-0.6R \le p_t^{\mathrm{Grid}}, q_t^{\mathrm{Grid}} \le R \tag{4d}$$

where $i \in N$ and $ij \in M$ in (2) and (4) while $i \in Ns$ in (3), and $t' = 1, \ldots, 24$. The meanings of symbols used in this section are listed in Table I. Unless otherwise stated, variable t belongs to set *T* throughout the paper. To maintain good power quality and encourage local balance of reactive power, some power utilities require that the reverse power flow (especially the reactive power flow) through the substation transformer should not exceed a certain percentage of the substation MVA capability, for instance 60% [19]. This requirement is reflected in (4d).

Constraints in (2) denote the branch flow constraints; (2a)–(2c) are linear while (2d) is quadratic; (3a)–(3c) represent the

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TABLE II
THE REFORMULATED DESOS MODELS

Notation	DESOS 1	DESOS 2	DESOS 3
Objective function	(1a)	(1b)	(5a)
Constraints	(2), (3a) and (4)	(2), (3a) - (3d) and (4)	(2), (3a)–(3d), (4) and (5b)

state and rate of charging/discharging constraints of the DES respectively [20]; (3d) captures the active power losses in the DES units; (3e) is a complementary constraint which is used to guarantee that at most one of $\bar{p}_{i,t}^{\mathrm{DCH}}$ and $\bar{p}_{i,t}^{\mathrm{CH}}$ is non-zero, and (4) denotes the system constraints. Note that, in the above model, one of (4a) and (4b) is redundant due to the relation described in (2d). However, the equality in (2d) may not hold in an optimal solution to a convex relaxation. Thus, both (4a) and (4b) are retained.

Objective function (1a) represents minimizing purchase cost of grid energy which is analogous to the objective function of minimizing generation cost in OPF problems. Objective functions (1b) and (1c) minimize network losses and voltage magnitude deviation respectively. These objective functions are frequently used in reactive power optimization problems [21] and play important roles in distribution system operations. In the variable space, (1a) and (1b) are monotonic over the feasible set while (1c) is not. Based on the operational requirement, only one of the functions in (1) will be chosen as the objective function of the required optimization model for the DESOS. The complementary constraint (3e) is a bilinear equality which is hard to convexify and solve. A solution strategy is developed and discussed in the next section to get rid of this constraint.

B. Convex Reformulation of Objective Function (1c)

Both (1a) and (1b) are linear, and (1c) which contains the absolute value sign is not. Therefore, before applying the convex relaxations to the DESOS problems, (1c) needs to be preprocessed to facilitate the convexification. Introducing auxiliary variables $u_{i,t}(i \in Ns)$ which are positive, (1c) can be reformulated as

$$\min f_3 = \sum_t^T \left(\sum_i^N u_{i,t} + \varepsilon \sum_i^{N_S} p_{i,t}^{\mathrm{DESloss}} \right) \quad (5a)$$

$$-u_{i,t} \le v_{i,t} - v_i^{\mathrm{set}} \le u_{i,t}, \quad (i \in N). \quad (5b)$$

where the quantity ϵ is an arbitrarily small positive value. Hence, three reformulated DESOS models for which the details are given in Table II are obtained. Constraint (3e) is a bilinear equality that is hard to convexify and solve. It is found that, by introducing the epsilon-term in (5a), constraint (3e) can be omitted. More detailed explanations are given in the following remarks.

For the DESOS models, note the following:

i) In any optimal solution of (*DESOS 2*), $p_{i,t}^{\mathrm{DCH}}$ and $p_{i,t}^{\mathrm{CH}}$ cannot be non-zero simultaneously even constraint (3e) is dropped. Suppose that $\bar{z}$ is an optimal solution to (*DESOS*) where both $\bar{p}_{i,t}^{\mathrm{DCH}}$ and $\bar{p}_{i,t}^{\mathrm{CH}}$ are non-zero and $\bar{p}_{i,t}^{\mathrm{DCH}} + \bar{p}_{i,t}^{\mathrm{CH}} = K > 0$ ($\bar{p}_{i,t}^{\mathrm{DCH}} + \bar{p}_{i,t}^{\mathrm{CH}} = K < 0$). There always exists another solution $\hat{z}$ where $\hat{p}_{i,t}^{\mathrm{DCH}} = K$ and $\hat{p}_{i,t}^{\mathrm{CH}} = 0$ ($\hat{p}_{i,t}^{\mathrm{DCH}} = 0$ and $\hat{p}_{i,t}^{\mathrm{CH}} = K$). It is easy to show that $\hat{z}$ satisfies (2a), (3a)–(3c), which means $\hat{z}$ is feasible to (DESOS), and $\hat{p}_{i,t}^{\mathrm{DESloss}} \le \bar{p}_{i,t}^{\mathrm{DESloss}}$ implying that $\hat{f}_2 \le \bar{f}_2$. Hence, $\hat{z}$ is the optimal solution rather than $\bar{z}$.

ii) In any optimal solution of (*DESOS 1*), $p_{i,t}^{\mathrm{DCH}}$ and $p_{i,t}^{\mathrm{CH}}$ also cannot be non-zero simultaneously. From i), it can be observed that the term $\sum_i^{N_S} p_{i,t}^{\mathrm{DESloss}}$ in the objective function forces at most one of $p_{i,t}^{\mathrm{DCH}}$ and $p_{i,t}^{\mathrm{CH}}$ to be non-zero in a specific time interval. Actually, within the feasible set, $\sum_i^{N_S} p_{i,t}^{\mathrm{DESloss}}$ is concealed in the objective function (1a), since the relation ($p_t^{\mathrm{Grid}} = \sum p_t^{\mathrm{Load}} + \sum p_t^{\mathrm{Loss}}$) holds for a feeder. In this paper, the optimization problems are solved based on given load profiles. As a result, the grid power is related to losses in the feeder according to the above relation. With the second term in (5a), this conclusion is also valid for (*DESOS 3*) no matter how small ϵ is.

iii) Objective function (5a) is equivalent to objective function (14) in [21] from an engineering point of view if ϵ is small enough. Therefore, one can choose an arbitrarily small value for ϵ so that (5a) is equivalent to (14) in [21] in engineering applications.

C. BSDP Relaxation

The SDP relaxation is a promising convex relaxation that has been successfully applied to some large-scale OPF problems [22]. It seems that the SDP-based methods are more adaptive to high-order nonlinear DES models [23]. Hence, the SDP-based relaxations are adopted to convexify the DESOS cases for which the SOCP relaxation is not exact or not applicable. Note that, in this paper, a convex relaxation for a non-convex problem is defined to be exact if the optimal solution obtained by solving the convexified problem is a feasible solution to the original non-convex problem. This subsection introduces the BSDP relaxation for the DESOS algorithm in radial networks based on the BFM.

Let $\boldsymbol{x}_t := [\boldsymbol{P}_{\mathrm{t}}^{\mathrm{T}} \boldsymbol{Q}_{\mathrm{t}}^{\mathrm{T}} \boldsymbol{v}_{\mathrm{t}}^{\mathrm{T}} \boldsymbol{\ell}_{\mathrm{t}}^{\mathrm{T}}]^{\mathrm{T}}$ for $t \in T$, the auxiliary semidefinite variable matrix $\boldsymbol{X}_t$ are defined in terms of $\boldsymbol{x}_t$. Replacing the quadratic terms with the corresponding entries of $\boldsymbol{X}_t$ yields the BSDP relaxation for DESOSs. As a result, (2d) and (4a) are replaced by the following linear constraints respectively

$$X_{(v_{i,t}\ell_{ij,t})} = X_{(P_{ij,t}^2)} + X_{(Q_{ij,t}^2)} \quad (6a)$$

$$X_{(P_{ij,t}^2)} + X_{(Q_{ij,t}^2)} \le \bar{S}_{ij}^2 \quad (6b)$$

where $i \in N, ij \in M$; $X_{(v_{i,t}\ell_{ij,t})}$ is the entry of $\boldsymbol{X}_t$ that corresponds to $v_{i,t}\ell_{ij,t}$ in the matrix $\boldsymbol{x}_t\boldsymbol{x}_t^{\mathrm{T}}$. Since the diagonal entries of $\boldsymbol{X}_t$ correspond to those of $\boldsymbol{x}_t\boldsymbol{x}_t^{\mathrm{T}}$ which are squared terms, it is easy to obtain the following bound constraints for them:

$$\left(\max\left\{0, \underline{x}_{i,t}\right\}\right)^2 \le X_{ii,t} \le \max\left\{\underline{x}_{i,t}^2, \overline{x}_{i,t}^2\right\} (i = 1, \ldots, n) \quad (6c)$$

where n is the number of variables in the vector $\boldsymbol{x}_{\mathrm{t}}$; $\underline{\boldsymbol{x}}_t = [\underline{\boldsymbol{P}}_t{}^{\mathrm{T}} \ \underline{\boldsymbol{Q}}_t{}^{\mathrm{T}} \ \underline{\boldsymbol{v}}_t{}^{\mathrm{T}} \ \underline{\boldsymbol{\ell}}_t{}^{\mathrm{T}}]^{\mathrm{T}}$ and $\bar{\boldsymbol{x}}_t = [\bar{\boldsymbol{P}}_t^{\mathrm{T}} \ \bar{\boldsymbol{Q}}_t^{\mathrm{T}} \ \bar{\boldsymbol{v}}_t^{\mathrm{T}} \ \bar{\boldsymbol{\ell}}_t^{\mathrm{T}}]^{\mathrm{T}}$. For sim-

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TABLE III
THE BSDP RELAXATION FOR DESOSS

Objective function	(1a), (1b) or (5a)
Constraints	(2a)–(2c), (3a)–(3d), (4b)–(4d), (6) and (5b), if (5a) is chosen as objective function

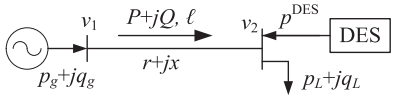


Fig. 1. Topology of the simple system.

plicity, $-\bar{S}_{ij}$ and $\bar{S}_{ij}$ can be used as the lower and upper bounds respectively for $\boldsymbol{P}_t^{\mathrm{T}}$ and $\boldsymbol{Q}_t^{\mathrm{T}}$.

Relaxing (2d) into (6a) has been considered in [10] and the Shor's inequality (6d) [24] is added to strengthen the SDP relaxation. The SDP relaxation for a conventional OPF problem in BFM studied in [10] is extended to DESOSs and regarded as the BSDP relaxation in this paper

$$\begin{bmatrix} 1 & \boldsymbol{x}_t^{\mathrm{T}} \\ \boldsymbol{x}_t & \boldsymbol{X}_t \end{bmatrix} \succeq 0. \tag{6d}$$

As a result, for DESOSs, the detailed model of the BSDP relaxation is tabulated in Table III.

III. IMPACT OF OBJECTIVE FUNCTIONS ON EFFICIENCY OF THE CONVEX RELAXATIONS

Illustrative examples are provided in [13], [14] and [17] to demonstrate the impact of some bounds/constraints on the exactness of the SOCP relaxation for OPF problems with a unique objective function. Inspired by these examples, this section studies the impact of objective functions on the efficiency of the convex relaxations for the DESOS problems through an illustrative example. In this example, we try to plot the feasible sets of the convex relaxations to provide an intuitive understanding. However, the semidefinite constraint (6d) in the BSDP relaxation is difficult to handle for this purpose. Hence, instead, the SOCP relaxation is studied in this illustrative example. The SOCP relaxation of the quadratic equality (2d) is given as

$$\left\| \begin{bmatrix} \sqrt{2}P_{ij,t} \\ \sqrt{2}Q_{ij,t} \\ v_{i,t} \\ \ell_{ij,t} \end{bmatrix} \right\|_2 \le \begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2}P_{ij,t} \\ \sqrt{2}Q_{ij,t} \\ v_{i,t} \\ \ell_{ij,t} \end{bmatrix}. \tag{7}$$

A. Illustrative Example

A 2-bus, 1-line system as shown in Fig. 1 is designed, where bus 1 is assumed to be an infinite bus. The variables and parameters of this simple system are defined in Table IV. For the sake of simplicity, the charging/discharging efficiency are considered to be 100% and the charging/discharging power are represented by one variable in this simple example.

TABLE IV
PARAMETERS OF THE 2-BUS, 1-LINE SYSTEM

Type	Symbol	Bounds/Values
Variable	ℓ	0 to 0.22 p.u.
Variable	v_2*	0.9 to 1.1 p.u.
Variable	p_g	–12 MW to 20 MW
Variable	q_g	–9 MVar to 15 MVar
Variable	P^{DES}	–10 MW to 10 MW
Parameter	v_1	$1.04^2 = 1.0816$ p.u.
Parameter	$r + jx$	$10 + j4$ ohm
Parameter	$p_L + jq_L$	$10 + j6$ MVA
Parameter	Nominal voltage	12.47 kV
Parameter	Power base	100 MVA

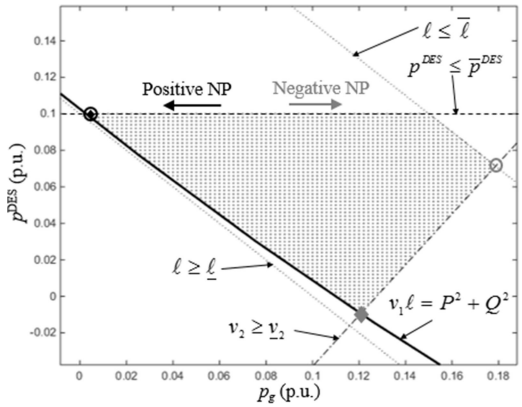


Fig. 2. The projection of the feasible set of (8) onto the (p_g, p^{DES})-space. When the NP is positive, the descent direction of objective function f_1 is denoted by the black arrow. In this case, the optimal solutions of the SOCP relaxation and the original problem are identical, which means the SOCP relaxation is exact. When the NP is negative, the SOCP relaxation of this case is not exact.

The branch AC power flow of this system is given as

$$\begin{cases} p_g = P, q_g = Q \\ p_L - p^{\mathrm{DES}} = P - r\ell, q_L = Q - x\ell \\ v_1 - v_2 = 2rP + 2xQ - \left(r^2 + x^2\right)\ell \\ P^2 + Q^2 = v_1\ell. \end{cases} \tag{8}$$

Eliminating the branch powers P and Q, the projections of the feasible set of (8) onto the (p_g, p^{DES}), (ℓ, p^{DES}) and (v_2, p^{DES}) spaces are shown in Figs. 2–4 respectively.

In the above figures, the feasible sets of the SOCP relaxation are described by the shaded regions. The black boundaries of these shaded regions are the feasible sets of the original DESOS problem. The curves are marked with the constraints they represent. The optimal solutions of the SOCP relaxation and the original problem are denoted by a circle and a solid diamond respectively in the figures. In Fig. 2, a case of negative nodal prices (NP) is taken into account since negative NPs may occur due to some reasons in actual power systems, for example the congestion on transmission lines.

B. Observations and Discussion

From this illustrative example, the following observations are made:

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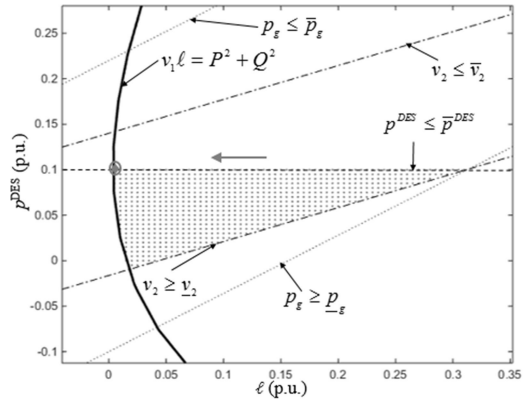


Fig. 3. The projection of the feasible set of (8) onto the (ℓ, p_{DES})-space. Since the charging/discharging efficiency of the DES are assumed to be 100% and no transformers are taken into account, the only term in objective function f_2 is $r\ell$ whose descent direction is denoted by the grey arrow. The SOCP relaxation of this case is exact.

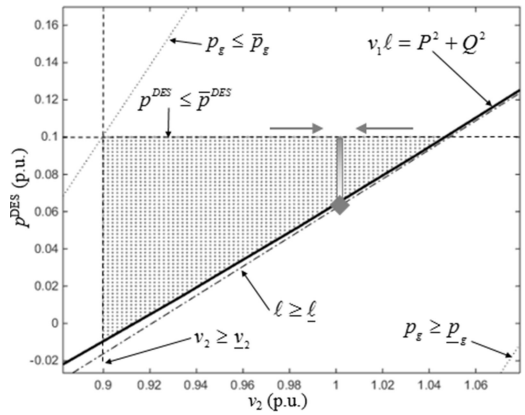


Fig. 4. The projection of the feasible set of (8) onto the (v_2, p^{DES})-space. Objective function f_3 is not monotonic over the feasible set, hence its descent directions are represented by the grey arrows. The optimal solution of the original problem is denoted by the grey diamond while that of the SOCP relaxation is not unique and denoted by the grey cylinder. Therefore, the SOCP relaxation of this case is most likely not exact.

a) The SOCP relaxation offers a convex superset a portion of whose boundaries is the feasible set of the original non-convex problem. An exact globally optimal solution of the original problem can be achieved if the objective function is monotonic over the feasible set and its descent direction points to this part of boundaries. For a given DESOS problem, there is a specific range of descent directions under which its SOCP relaxation is exact. A mathematical description of such a range of descent directions is given in [5, Proposition 1].
b) For an objective function which does not satisfy [5, Proposition 1], an unsatisfactory solution may result, e.g. the negative NP case in Fig. 2. The bounds of some variables also affect the exactness of the SOCP relaxation, for more details please refer to [17]. In fact, the state of charging/discharging constraint (3a) may also affect the exactness of the SOCP relaxation for the DESOS problems.
c) The feasible set of the original non-convex problem is not necessarily a part of the boundaries of the convex superset offered by the BSDP relaxation. As shown in [10], the SOCP relaxation of the branch AC power flow equalities (2d) (as shown in (7)) is tighter than the BSDP relaxation ((6a) and (6d)). Consequently, for some potential objective functions of the DESOS problems for which the SOCP relaxation does not work well, the BSDP relaxation may also be ineffective. Alternative convex relaxations are required to achieve a better solution when these objective functions are adopted to meet some specific operational requirements.

IV. Non-Iterative ESDP Relaxations

In this section, some convex constraints which can be used to approximate the rank-1 constraint $\boldsymbol{X} = \boldsymbol{x}\boldsymbol{x}^{\mathrm{T}}$ are introduced to construct ESDP relaxations in the non-iterative framework for the DESOS problems. One primary objective of this paper is to obtain, through numerical comparison, tighter but computationally effective non-iterative ESDP relaxations for the DESOS problems.

A. Rank-2 Linear Inequalities

The constraints in (4b) and (4c) can be reformulated as

$$\begin{cases} v_{i,t} - \underline{v}_i \ge 0 \\ \bar{v}_i - v_{i,t} \ge 0 \\ \ell_{ij,t} - 0 \ge 0 \\ \bar{\ell}_{ij} - \ell_{ij,t} \ge 0 \end{cases} (i \in N, ij \in M) \tag{9a}$$

which are linear inequality constraints on the variables in $\boldsymbol{x}_t$ only. An example of pairwise products of the linear inequalities in (4b) and (4c) is

$$\begin{cases} (v_{i,t} - \underline{v}_i)\,\ell_{ij,t} \ge 0 \\ (v_{i,t} - \underline{v}_i)\left(\bar{\ell}_{ij} - \ell_{ij,t}\right) \ge 0 \\ (\bar{v}_i - v_{i,t})\,\ell_{ij,t} \ge 0 \\ (\bar{v}_i - v_{i,t})\left(\bar{\ell}_{ij} - \ell_{ij,t}\right) \ge 0 \end{cases} (i \in N, ij \in M). \tag{9b}$$

The polynomial expansions of (9b) are the well-known McCormick inequalities [25]. Replacing the quadratic terms in (9b) with the corresponding entries in $\boldsymbol{X}_t$ results in the following rank-2 linear inequalities (R2LI) [26]

$$\begin{cases} \boldsymbol{X}_{(v_{i,t}\ell_{ij,t})} - \underline{v}_i\ell_{ij,t} \ge 0 \\ \boldsymbol{X}_{(v_{i,t}\ell_{ij,t})} - \bar{\ell}_{ij}v_{i,t} - \underline{v}_i\ell_{ij,t} + \underline{v}_i\bar{\ell}_{ij} \le 0 \\ \boldsymbol{X}_{(v_{i,t}\ell_{ij,t})} - \bar{v}_i\ell_{ij,t} \le 0 \\ \boldsymbol{X}_{(v_{i,t}\ell_{ij,t})} - \bar{\ell}_{ij}v_{i,t} - \bar{v}_i\ell_{ij,t} + \bar{v}_i\bar{\ell}_{ij} \ge 0. \end{cases} (i \in N, ij \in M) \tag{9c}$$

Since the R2LIs in (9c) stem from pairwise products of the bound constraints which can be regarded as a special form of linear inequalities, they are usually referred to as the "Reformulation-linearization technique" (RLT) [27]. Constraint (6c) is widely adopted to provide valid bounds on the diagonal entries of $\boldsymbol{X}_t$ in literature. The RLT itself is a convex relaxation of a quadratically constrained quadratic programming (QCQP) problem. The matrix $\boldsymbol{X}_t$ is not required to be positive semidefinite in a pure RLT relaxation. Note that, the case given in (9c) is just an example. Similar constraints can be obtained for all the entries of $\boldsymbol{X}_t$, e.g. $X_{(v_{i,t}v_{j,t})}(i, j \in N)$ and $X_{(\ell_{ij,t}\ell_{kh,t})}(ij, kh \in M)$, in the same way. Therefore, the RLT is considered in this paper for tightening the BSDP relaxation of

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DESOSs. Since the RLT offers valid bounds on both the diagonal and non-diagonal entries of $\boldsymbol{X}_t$, it can effectively strengthen the SDP relaxation. In the BFM used in this paper, $v_{i,t}$ and $\ell_{ij,t}$ are non-negative variables. It is easy to verify that (6c) corresponding to these variables are redundant when the RLT is used, since (9c) on the diagonal entries of $\boldsymbol{X}_t$ are tighter than (6c).

If there exist general linear inequalities which can be expressed as

$$\boldsymbol{c}_i^{\mathrm{T}}\boldsymbol{x}_t - b_i \geq 0 \qquad (i \in S_{LI}) \tag{9d}$$

the following general R2LIs can be obtained

$$\begin{cases} \boldsymbol{X}\boldsymbol{c}_i - b_i\boldsymbol{x} + b_i\underline{\boldsymbol{x}} - \underline{\boldsymbol{x}}\boldsymbol{x}^{\mathrm{T}}\boldsymbol{c}_i \leq 0 \\ \boldsymbol{X}\boldsymbol{c}_i - b_i\boldsymbol{x} + b_i\bar{\boldsymbol{x}} - \bar{\boldsymbol{x}}\boldsymbol{x}^{\mathrm{T}}\boldsymbol{c}_i \geq 0 \\ \boldsymbol{c}_i^{\mathrm{T}}\boldsymbol{X}\boldsymbol{c}_j - \left(b_i\boldsymbol{c}_j^{\mathrm{T}} + b_j\boldsymbol{c}_i^{\mathrm{T}}\right)\boldsymbol{x} + b_ib_j \geq 0 \end{cases} \qquad (i,j \in S_{\mathrm{LI}}) \tag{9e}$$

where S_{LI} is the set of linear inequality constraints on $\mathbf{x}_t$ only; the first inequality in (9e) originates from the product of the lower bound of $\mathbf{x}_t$ and (9d) while the second one comes from that of the upper bound of $\mathbf{x}_t$; and, the third one stems from self-product of two inequalities in (9d).

B. Valid Linear Equalities

Constraints (2a) and (2b) for the buses which are not the substation bus or the DES-connected buses, and (2c) can be formulated as the following general linear equalities

$$\boldsymbol{c}_i^{\mathrm{T}}\boldsymbol{x}_t = b_i \,(i \in S_{\mathrm{LE}}) \tag{10a}$$

where S_{LE} is the set of linear equality constraints on $\boldsymbol{x}_t$ only. Pre-multiplying the linear equalities in (10a) by $\boldsymbol{x}_t$ and replacing the quadratic terms with $\boldsymbol{X}_t$ results in

$$\boldsymbol{X}_t\boldsymbol{c}_i = b_i\boldsymbol{x}_t \quad (i \in S_{\mathrm{LE}}). \tag{10b}$$

Replacing the quadratic terms in the pairwise products of the linear equalities in (10a) with $\boldsymbol{X}_t$, the following valid linear equalities (VLE) are obtained

$$\boldsymbol{c}_i^{\mathrm{T}}\boldsymbol{X}_t\boldsymbol{c}_j - \left(b_i\boldsymbol{c}_j^{\mathrm{T}} + b_j\boldsymbol{c}_i^{\mathrm{T}}\right)\boldsymbol{x}_t + b_ib_j = 0 \;(i,j \in S_{\mathrm{LE}}) \tag{10c}$$

$$\boldsymbol{c}_i^{\mathrm{T}}\boldsymbol{X}_t\boldsymbol{c}_j = b_ib_j \;(i,j \in S_{\mathrm{LE}}). \tag{10d}$$

Linear equalities (10b)–(10d) have been used to tighten the SDP relaxation for small-scale general QCQP problems which contain linear equality constraints [28].

Proposition 1: Constraint (10c) is equivalent to (10d) and tighter than (10b). (10a) is redundant when (10c) is used to tighten the BSDP relaxation.

Proof: It is easy to verify that (10d) are equivalent to (10b) when $i = j$. hence, (10d) is a subset of (10b) which means (10d) is tighter than (10b). By substituting (10a) into (10c), it suffices to show that (10c) is equivalent to (10d). When $i = j$, constraint (10c) becomes $\boldsymbol{c}_i^T\boldsymbol{X}_t\boldsymbol{c}_i - 2\mathrm{b_i}\boldsymbol{c}_\mathrm{i}^{\mathrm{T}}\boldsymbol{x}_\mathrm{t} + \mathrm{b_i^2} = 0$. Due to the Shor's inequality in (6d), $0 = \boldsymbol{c}_\mathrm{i}^{\mathrm{T}}\boldsymbol{X}_\mathrm{t}\boldsymbol{c}_\mathrm{i} - 2\mathrm{b_i}\boldsymbol{c}_\mathrm{i}^{\mathrm{T}}\boldsymbol{x}_\mathrm{t} + \mathrm{b_i^2} \geq \boldsymbol{c}_\mathrm{i}^{\mathrm{T}}\boldsymbol{x}_\mathrm{t}\boldsymbol{x}_\mathrm{t}^{\mathrm{T}}\boldsymbol{c}_\mathrm{i} - 2\mathrm{b_i}\boldsymbol{c}_\mathrm{i}^{\mathrm{T}}\boldsymbol{x}_\mathrm{t} + \mathrm{b_i^2} = \left(\boldsymbol{c}_\mathrm{i}^{\mathrm{T}}\boldsymbol{x}_\mathrm{t} - \mathrm{b_i}\right)^2$, which means $\boldsymbol{c}_i^T\boldsymbol{x}_t - \mathrm{b_i} = 0$. As a result, (10a) is redundant. □

Proposition 1 implies that (10c) is preferred for obtaining tight ESDP relaxations for DESOS problems.

C. Semidefinite Inequality_1

Based on the Schur complement condition for positive definiteness [29], (4a) can be further reformulated as

$$\begin{bmatrix} \boldsymbol{I}_2 & & P_{ij,t} \\ & & Q_{ij,t} \\ P_{ij,t} & Q_{ij,t} & \bar{S}_{ij}^2 \end{bmatrix} \succeq 0, \; (ij \in M) \tag{11}$$

where $\boldsymbol{I}_2$ is a 2×2 identity matrix.

Imposing (11) on the BSDP relaxation will most likely yield a stronger SDP relaxation for the DESOS problems. Because (11) is the semidefinite expression of (4a) and combining (11) and (6b) provides a link between $P_{ij,t}^2 + Q_{ij,t}^2$ and $X_{(P_{ij,t}^2)} + X_{(Q_{ij,t}^2)}$, in other words a link between $\mathrm{tr}(\boldsymbol{A}_{ij,t}\boldsymbol{x}_t\boldsymbol{x}_t^T)$ and $\mathrm{tr}(\boldsymbol{A}_{ij,t}\boldsymbol{X}_t)$, where $\boldsymbol{A}_{ij,t}$ is an $n \times n$ matrix and the only non-zero entries of $\boldsymbol{A}_{ij,t}$ are the two diagonal entries corresponding to $P_{ij,t}$ and $Q_{ij,t}$, which equal to 1.

D. Semidefinite Inequality_2

Multiplying (11) by the first term of (9a) for instance, the following semidefinite inequality is obtained

$$\begin{bmatrix} v_{i,t} - \underline{v}_i & 0 & P_{ij,t}v_{i,t} - P_{ij,t}\underline{v}_i \\ 0 & v_{i,t} - \underline{v}_i & Q_{ij,t}v_{i,t} - Q_{ij,t}\underline{v}_i \\ P_{ij,t}v_{i,t} - P_{ij,t}\underline{v}_i & Q_{ij,t}v_{i,t} - Q_{ij,t}\underline{v}_i & \bar{S}_{ij}^2 v_{i,t} - \bar{S}_{ij}^2\underline{v}_i \end{bmatrix} \succ 0. \quad (i \in N, ij \in M) \tag{12a}$$

Replacing the quadratic terms in (12a) with the corresponding entries of $\boldsymbol{X}_t$ yields 12(b) as shown at the bottom of the page.

For a general description of the semidefinite inequalities in (12b), please refer to [30]. In fact, inequalities (12a) are implicitly contained in the DESOS models since a given $(P_{ij,t}, Q_{ij,t}, v_{i,t})$ that satisfies (4a) and (4c) will also satisfy (12a). As a result, constraint (12b) in conjunction with (4c) and (11) which is equivalent to (4a) offers approximations to the relations that $X_{(Pij,tvi,t)} = P_{ij,t}v_{i,t}$ and $X_{(Qij,tvi,t)} = Q_{ij,t}v_{i,t}$ which are weak in the BSDP relaxation.

$$\begin{bmatrix} v_{i,t} - \underline{v}_i & 0 & X_{(P_{ij,t}v_{i,t})} - P_{ij,t}\underline{v}_i \\ 0 & v_{i,t} - \underline{v}_i & X_{(Q_{ij,t}v_{i,t})} - Q_{ij,t}\underline{v}_i \\ X_{(P_{ij,t}v_{i,t})} - P_{ij,t}\underline{v}_i & X_{(Q_{ij,t}v_{i,t})} - Q_{ij,t}\underline{v}_i & \bar{S}_{ij}^2 v_{i,t} - \bar{S}_{ij}^2\underline{v}_i \end{bmatrix} \succeq 0 \quad (i \in N,\, ij \in M) \tag{12b}$$

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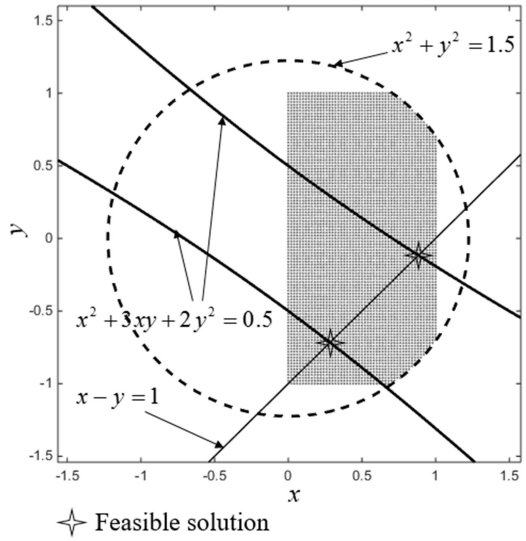


Fig. 5. Feasible set of system (13a).

The left hand side of (12b) includes 3×3-dimensional matrices which make constraints (12b) easy to handle computationally. Moreover, formulating constraints (11) and (12b) in the second-order cone form in the computer program will probably improve the computational efficiency. Note that each term in (9a) can be multiplied by (11) to generate a semidefinite inequality which is similar to (12b). Therefore, the complete form of (12b) can impact every entry of $\boldsymbol{X}_t$, by which one can expect a tight SDP relaxation with (12b) imposed.

E. Geometric Interpretation

To provide a geometric interpretation of the convex constraints introduced above, the following 2-dimensional system, whose feasible set is given in Fig. 5, is studied in this subsection. Note that the feasible set of the SDP relaxation for a system whose dimension is higher than 2 is hard to visualize

$$\begin{cases} x^2+3xy+2y^2=0.5 \\ x-y=1 \\ x^2+y^2 \le 1.5 \\ 0 \le x \le 1 \\ -1 \le y \le 1. \end{cases} \tag{13a}$$

The BSDP relaxation of system (13a) is

$$\begin{cases} X_{11}+3X_{12}+2X_{22}=0.5 \\ x-y=1 \\ X_{11}+X_{22} \le 1.5 \\ 0 \le X_{11}, X_{22} \le 1 \\ 0 \le x \le 1, -1 \le y \le 1 \\ \boldsymbol{X}=\begin{bmatrix} X_{11} & X_{12} \\ X_{12} & X_{22} \end{bmatrix} \succeq 0. \end{cases} \tag{13b}$$

whose feasible region is described in Fig. 6.

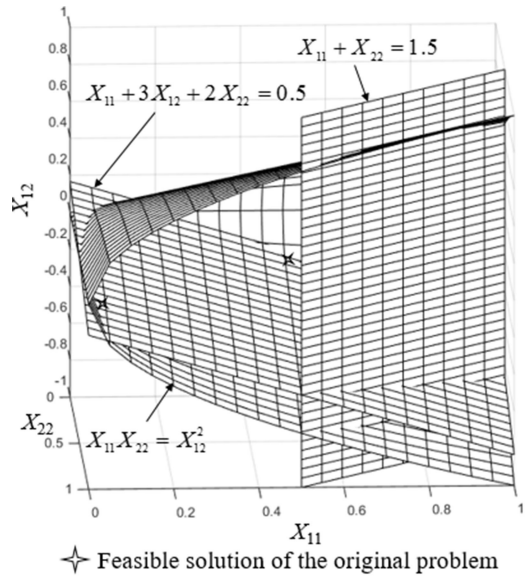


Fig. 6. Feasible set of system (13b) in the (X_{11}, X_{22}, X_{12})-space. The feasible set of (13b) is the region on the plane $X_{11}+3X_{12}+2X_{22}=0.5$ that is enclosed by the surface $X_{11}X_{22}=X_{12}^2$ and the plane $X_{11}+X_{22}=1.5$ within the bound constraints.

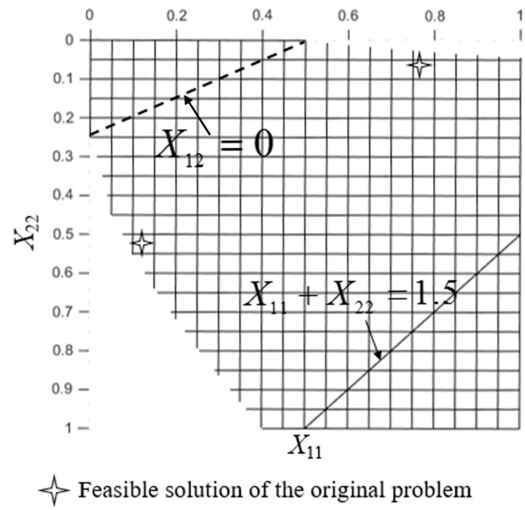


Fig. 7. Geometric interpretation of RLT. The projection of the feasible set of (13b) onto the (X_{11}, X_{22})-plane is the meshed region to the left of line $X_{11}+X_{22}=1.5$. The dashed line is the projection of the intersection of $X_{11}+3X_{12}+2X_{22}=0.5$ and $X_{12}=0$ onto the (X_{11}, X_{22})-plane. The triangular area to the left of the dashed line is cut off when $X_{12} \le 0$ is added.

Following the steps introduced in Section IV-A, the following RLT constraints for this system are obtained

$$\begin{cases} \max\{2x-1, 0\} \le X_{11} \le x \\ \max\{2y-1, -2y-1\} \le X_{22} \le 1 \\ \max\{x+y-1, -x\} \le X_{12} \le 0. \end{cases} \tag{13c}$$

The first constraint in (13c) demonstrates that, for a nonnegative variable, the RLT constraint on the diagonal entry of $\boldsymbol{X}$ corresponding to this variable dominates that in (6c). The efficiency of the bound constraint $X_{12} \le 0$ which is one of the RLT constraints in (13c) is illustrated in Fig. 7.

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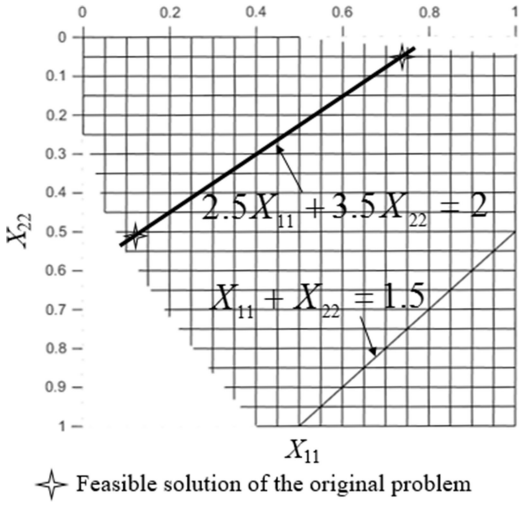


Fig. 8. Geometric interpretation of VLE. The line segment $2.5X_{11}+3.5X_{22} = 2$ is the projection of the intersection of $X_{11}+3X_{12}+2X_{22} = \$\,0.5$ and (13e) onto the (X_{11}, X_{22})-plane within the feasible set. This line segment is exactly the resulting feasible set after imposing (13e) on system (13b).

Based on the linear equality $x - y = 1$, the following VLE (10b) is obtained for this system

$$\begin{bmatrix} X_{11} & X_{12} \\ X_{12} & X_{22} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \times \begin{bmatrix} x \\ y \end{bmatrix} \tag{13d}$$

which is equivalent to (combining $x - y = 1$)

$$X_{11} - 2X_{12} + X_{22} = 1. \tag{13e}$$

The feasible set of the SDP relaxation of (13a) with (13e) imposed is given in Fig. 8, which becomes as tight as a line segment. This example numerically demonstrates that the VLE is effective if there exists a considerable number of linear equality constraints in the original problem. Another important reason why the BFM is preferred to formulate the AC power flow is that most constraints in the BFM are linear equalities.

Based on the convex quadratic constraint in (13a) and (11), the semidefinite inequality in (13f) is obtained. Since the link between $\boldsymbol{X}$ and $[x, y]$ is very weak in the BSDP relaxation (13b), it is hard to obtain the geometric figure for semidefinite inequality_1. However, the efficiency of the SI_1 has been verified by numerical studies in power systems given in Section V

$$\begin{bmatrix} 1 & 0 & x \\ 0 & 1 & y \\ x & y & 1.5 \end{bmatrix} \succeq 0. \tag{13f}$$

Multiplying (13f) by the lower bound constraint of x and replacing the quadratic terms with the corresponding entries in $\boldsymbol{X}$ yields the following semidefinite inequality

$$\begin{bmatrix} x & 0 & X_{11} \\ 0 & x & X_{12} \\ X_{11} & X_{12} & 1.5x \end{bmatrix} \succeq 0 \tag{13g}$$

which implies that

$$X_{11}^2 + X_{12}^2 \leq 1.5x^2 \leq 1.5. \tag{13h}$$

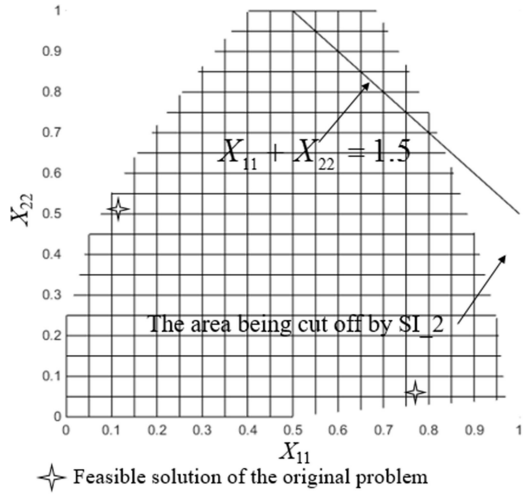


Fig. 9. Geometric interpretation of SI_2.

TABLE V
THE ESDP RELAXATIONS OF DESOSS

ESDP#	Constraints
1	(9c) for all the entries in $\boldsymbol{X}_t$ and (BSDP) in Table III with (6c) on $X_{(v_{i,t}^2)}$,$X_{(\ell_{ij,t}^2)}(i \in N, ij \in M)$ being eliminated.
2	(10c) and (BSDP) with constraints (2a) and (2b) for the buses which are not the substation bus or the DES-connected buses, and (2c) being eliminated
3	(BSDP) and (11)
4	(BSDP), (11) and (12b)
5	(10c), (11) and (BSDP) with constraints (2a) and (2b) for the buses which are not the substation bus or the DES-connected buses, and (2c) being eliminated
6	(9c), (10c) and (BSDP) with constraints (2a) and (2b) for the buses which are not the substation bus or the DES-connected buses, (2c) and (6c) on $X_{(v_{i,t}^2)}$,$X_{(\ell_{ij,t}^2)}(i \in N, ij \in M)$ being eliminated

Adding (13h) to the BSDP relaxation (13b), the resulting feasible set is plotted in Fig. 9. A portion of the feasible set of (13b) is cut off by (13h). The SI_2 constraint in (13g) will cut off a bigger portion of the feasible set since (13h) is looser than (13g).

F. Non-Iterative ESDP relaxations

The non-iterative ESDP relaxations are constructed by imposing one or some of the above convex constraints on the BSDP relaxation in Table III. Six ESDP relaxations are designed of which the details are given in Table V. Even in an iterative framework, the computations can also benefit from the non-iterative constraints since they are valid for the sub-problem at each iteration. This merit motivates researchers to develop techniques to lift valid cuts obtained at a node of the branch and bound tree to the entire tree [31]. Note that, in ESDP#3, the solution time may slightly benefit from replacing (11) with (4a) or its SOC form.

V. CASE STUDY

A. Test-Bed Systems and Case Design

The convex relaxations of the DESOS algorithm are tested on the IEEE 13, 37, 123-bus feeders [32] assuming that there

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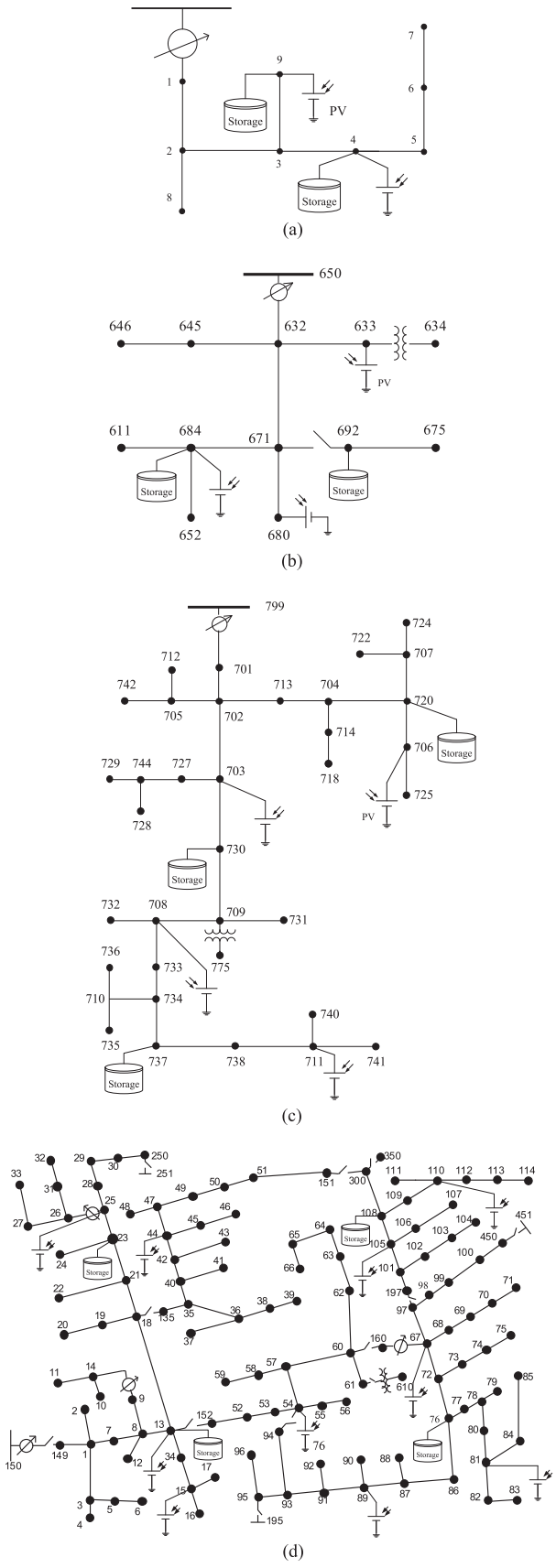


Fig. 10. Topologies of the test systems. (a) The 9-bus real world feeder in Arizona. (b) IEEE 13-bus feeder. (c) IEEE 37-bus feeder. (d) IEEE 123-bus feeder.

TABLE VI
PV SYSTEM AND DES LOCATION AND CAPACITY

Test system	PV location (bus #) and capacity	penetration
9-bus	4 (0.85 MW), 9 (0.65 MW)	30.5%
13-bus	633 (0.5 MW), 680 (0.2 MW), 684 (0.5 MW)	36.7%
37-bus	703 (0.3 MW), 706 (0.3 MW), 708 (0.3 MW), 711 (0.3 MW)	48.8%
123-bus	8, 15, 25, 44, 54, 67, 81, 89, 105, 110 (all PV systems have the same size of 0.2 MW)	57.3%
Test system	**DES location (bus #) and capacity**	
9-bus	4 (0.4 MW, 2 MWh), 9 (0.3 MW, 1.5 MWh)	
13-bus	684 (0.6 MW, 2.4 MWh), 692 (0.8 MW, 3.2 MWh)	
37-bus	720 (0.3 MW, 1.2 MWh), 730 (0.3 MW, 1.2 MWh), 737 (0.3 MW, 1.2 MWh)	
123-bus	13 (0.3 MW, 1.2 MWh), 23 (0.3 MW, 1.2 MWh), 76 (0.3 MW, 1.2 MWh), 108 (0.3 MW, 1.2 MWh)	

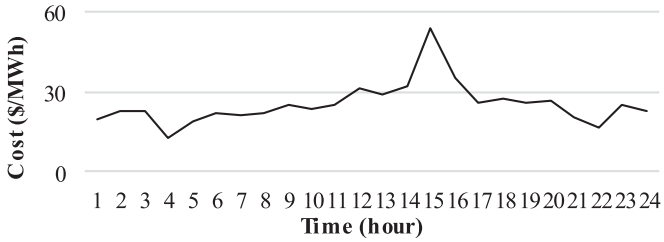


Fig. 11. The 24-hour NP curve for the 9-bus feeder case.

is high penetration of PV resources and a 9-bus real-world feeder in Arizona [5] respectively. The three-phase topologies of the three networks are shown in Fig. 10. The capacities of both PV and DES systems for all the test systems are listed in Table VI. The problems are solved by the solver MOSEK (version 7.1.0.34) [33] through the MATLAB toolbox YALMIP [34]. A computer with a 64-bit Intel i5-3230M dual core CPU at 2.60 GHz and 4 GB of RAM was used to run the test cases.

Multi-period demand and PV generation profiles are available only for the real-world feeder. Hence, in the real-world feeder cases, 24-hour data is used. The DES dynamic constraint (3a) is omitted and only a snapshot power flow is considered in each of the IEEE feeder cases. When objective function 1 is chosen, an actual 24-hour NP curve (as shown in Fig. 11) from the website of the ISO New England is used. The coefficient c in objective function 1 is set to be –30 $ /MWh for the IEEE cases. For all cases, η_c and η_d of the DES units are set to be 90% and 95% respectively. The ϵ in objective function (5a) is set to be 0.0001 in the test cases.

The optimal objective values (OOV) of the solutions are compared to quantify the tightness of the convex relaxations as researchers did in literature. As reported widely in literature, the SOCP relaxation outperforms the SDP relaxation in terms of computational efficiency. There is no surprise that the CPU times of solving the SOCP relaxation for the DESOS problems are less than those of the SDP relaxations. This paper aims at finding computationally inexpensive convex constraints that can tighten the BSDP relaxation. The proposed approaches (ESDP relaxations #1–6) are compared with the SOCP, BSDP relaxations and a nonlinear solver. The results are tabulated in Table VII.

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TABLE VII
RESULTS OF CASE STUDY

System	Relaxation	Objective Function 1 OOV ($)	Objective Function 1 CPU time (s)	Objective Function 2 OOV (p.u.)	Objective Function 2 CPU time (s)	Objective Function 3 OOV (p.u.)	Objective Function 3 CPU time (s)
9-bus real-world feeder	SOCP	1231.2	0.84	0.5832*	0.44	13.5148	0.50
	BSDP	1230.5	3.28	0.2822	2.68	13.4025	4.04
	ESDP1	1236.0	11.25	0.5769	5.82	13.4025	14.63
	ESDP2	1236.1	3.58	0.3350	2.86	14.2875	5.56
	ESDP3	1231.4	3.68	0.3369	3.21	13.4267	4.12
	ESDP4	1244.6	6.05	0.4235	4.25	13.4394	7.45
	ESDP5	1236.8	4.66	0.3350	3.46	14.4720	6.24
	ESDP6	1240.9	12.33	0.5792	6.62	14.2876	15.89
	IPOPT	1263.5	1.551	–	–	16.7597	2.424
IEEE 13-bus feeder	SOCP	-28.887	0.28	0.1276*	0.39	0.3933	0.36
	BSDP	-29.331	1.31	0.0485	1.40	0.3752	1.43
	ESDP1	-28.188	3.58	0.1201	3.75	0.4040	3.87
	ESDP2	-27.708	1.73	0.0558	1.53	0.4347	1.69
	ESDP3	-27.786	1.36	0.0534	1.40	0.4248	1.52
	ESDP4	-26.372	2.12	0.0597	2.13	0.4418	2.08
	ESDP5	-27.708	1.89	0.0565	1.74	0.4347	1.88
	ESDP6	-27.381	3.26	0.1256	3.39	0.4349	3.90
	IPOPT	-22.449	0.103	–	–	0.4598	0.126
IEEE 37-bus feeder	SOCP	-71.727	0.45s	0.0469*	0.47s	0.6206	0.30s
	BSDP	-72.213	6.81	0.0098	6.23	0.6039	7.17
	ESDP1	-72.192	24.02	0.0455	17.91	0.6206	22.13
	ESDP2	-69.303	12.69	0.0153	7.38	0.6165	13.84
	ESDP3	-70.421	7.60	0.0118	6.27	0.6239	7.45
	ESDP4	-66.697	8.74	0.0143	8.10	0.6559	10.60
	ESDP5	-69.303	13.02	0.0159	7.63	0.6248	14.67
	ESDP6	-68.670	19.22	0.0469	13.06	0.6394	25.90
	IPOPT	-66.288	0.134s	–	–	1.1307	0.132s
IEEE 123-bus feeder	SOCP	-43.851	0.58s	0.0306*	0.52s	2.9345	0.56s
	BSDP	-44.910	110.49	0.0106	103.93	2.7832	118.69
	ESDP1	-43.857	273.90	0.0296	315.32	2.9313	375.89
	ESDP2	-43.596	149.27	0.0145	129.91	3.0272	165.34
	ESDP3	-42.594	145.74	0.0127	118.06	3.1559	151.21
	ESDP4	-41.988	186.84	0.0140	175.95	3.2576	230.26
	ESDP5	-42.375	152.81	0.0152	141.76	3.1567	179.46
	ESDP6	-43.509	305.62	0.0305	328.31	3.1348	390.02
	IPOPT	–	–	–	–	–	–

Remark

a) In the IEEE feeder cases which are single-period cases, when objective function 1 with positive NPs is selected, the SOCP relaxation is exact. In the cases with negative NPs, the SOCP relaxation is not exact.

b) Several 24-hour NP curves (all positive) have been tested on the 9-bus feeder. The SOCP relaxation for the cases with some NP curves is exact, while for the other cases it is not. Generally, with objective function 1, the SOCP relaxation is only occasionally exact even when all values on the NP curves are positive.

c) When objective function 2 is chosen, the SOCP relaxation is in general exact. Because, minimizing $r\ell$ usually leads to a situation that constraint (7) is binding, which means (2d) is satisfied. However, it does not necessarily mean the SOCP relaxation is exact for all the DESOS cases where objective function 2 is adopted.

d) The SOCP relaxation is generally not exact when objective function 3 is selected, since it is strictly non-monotonic in the feasible set.

e) Dropping constraint (3a), the DESOS with objective function 1 and the NP curve in Fig. 6 is decomposed into 24 OPF problems. The SCOP relaxation for these 24 OPFs is exact. However, it is not for the 24-hour DESOS problem.

f) The exactness of the SCOP relaxation is determined using an index introduced in [5].

*The SOCP relaxation for these cases is exact.

B. Observations and Discussions

From the results shown in Table VII, one can observe that:

1) The exactness of the SOCP relaxation for the DESOS problems is sensitive to the descent direction of the chosen objective function. Moreover, it is also sensitive to some settings, for instance some bound values (see [17, Sections III and IV]).
2) Adding a linear constraint (e.g. the state of charging/discharging constraint (3a)) to an OPF problem may change the exactness of the SOCP relaxation for this problem. The effect of adding a linear constraint on exactness of the SOCP relaxation is similar to that of changing some bounds as shown in [17]. Please refer to Remark (e) in Table VII for more details.
3) The statements in observations 1) and 2) actually reveal several aspects that may determine whether a DESOS problem satisfies the conditions in [5, Proposition 1], i.e. the conditions for exactness of the SOCP relaxation, or not. It implies that, on the other hand, the exactness of the SOCP relaxation for the DESOSs is determined by various factors.
4) The ESDP relaxations, especially ESDP#4 and ESDP#5, are attractive alternatives for the cases where the SOCP relaxation is not exact, since one can usually find an ESDP relaxation that can provide a much tighter solution than SOCP relaxation does in these cases. The convex constraints introduced in Section IV do not considerably increase the computational burden of the BSDP relaxation except for the RLT.
5) Reducing the complexity of the BSDP relaxation is an interesting research topic that can be found in [22] and the references therein. It has been reported in [22] that the solution time for solving the BSDP relaxation in an OPF problem for the IEEE 300-bus system has been reduced to 3.7 s by exploiting the sparsity. This makes the ESDP relaxations valuable in application. Since the primary concern of this paper is improving the tightness of the convex relaxation under different objective functions, the SDP relaxations are computed in their original form without applying any complexity-reduction technique. Therefore, the dimension of the $\boldsymbol{X}_t$ matrix in the IEEE 123-bus feeder cases is 489×489, which results in high CPU times. The network structures of the radial systems studied in this

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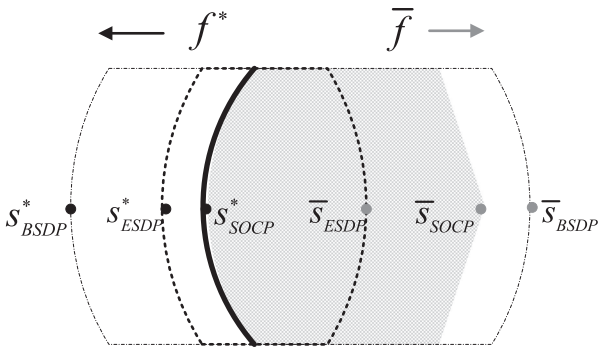


Fig. 12. Hypothesis about the tightness of the convex relaxations for the DESOSs in radial networks. Let $\Omega(\cdot)$ denotes a feasible set. Ω(DESOS) is denoted by the black thick curve. Ω(SOCP) is the shaded region while Ω(BSDP) and Ω(ESDP) are the regions enclosed by the dotted-dashed curve and the dotted curve respectively.

paper are very sparse. By exploiting the sparsity, the runtime of solving the ESDP relaxations for this systems can be significantly reduced.

6) For a number of DESOS cases, exactness of the convex relaxations still cannot be guaranteed. For practical purpose, some nonlinear solvers like IPOPT [35] are suggested to provide feasible solutions for these cases. Please refer to Table VII. Note that, IPOPT is failed to solve the IEEE 123-bus feeder cases.

Combining the results obtained in this paper and those reported in literature, a hypothesis on the tightness of the SOCP, BSDP as well as the ESDP relaxations for the DESOS problems in radial networks is illustrated in Fig. 12.

VI. Conclusion and Future Work

This paper found that, when the exactness of a convex relaxation for a given optimization problem is discussed, at least two elements should be considered: the tightness of the relaxation and the descent direction of the objective function. Some characteristics distinguish the DESOS problems examined in this paper from various OPF problems in literature. The widely-used SOCP relaxation is not exact and provides unsatisfactory solutions for a number of DESOS cases. In some cases where detailed battery and converter models are used, the SOCP relaxation may not be applicable. As an alternative, this paper introduces some convex constraints to strengthen the SDP relaxation for these DESOS cases in the non-iterative framework. In the DESOS cases where the SOCP relaxation is not exact, one can usually find an ESDP relaxation that can yield a tighter solution than the SOCP relaxation does. Hopefully, with some existing dimension-reduction techniques, the computational burden of the ESDP relaxations can be reduced to the level which is comparable to that of the SOCP relaxation. The ESDP relaxations are potential alternative solutions to the convexification of some QCQP problems in power systems like OPF. In future work, sparsity of radial networks will be explored to improve the computational efficiency of the ESDP relaxations.

Acknowledgment

The authors would like to thank Dr. H. Mittelmann of Arizona State University for the helpful suggestions to improve this paper.

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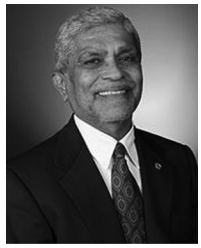
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